

EBN 111 Practical 1: Tutorial

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This tutorial serves as a guideline to installing and using the OrCAD Capture CIS and associated PSpice software.

If you do not have access to a personal computer running Microsoft Windows or prefer to work on the campus computers the required software is available on the computers in any of the engineering labs and you can skip to step 3.

1 Installing OrCAD

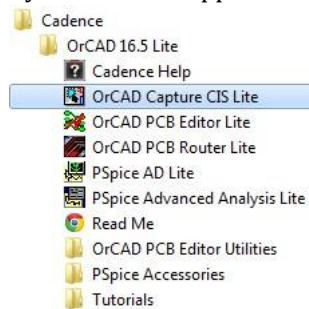
1. Obtain the OrCAD Demo package. It can be downloaded from the following link: http://ftp.ee.up.ac.za/pub/windows/eda/OrCAD_16.6_Full_Demo.zip.

It is highly recommended that you download the software from a computer on campus, as the file is approximately 1200MB in size.

2. Extract the archive, and install the software using setup.exe which can be located in the Disk1 or Lite_All directory. This will take some time.

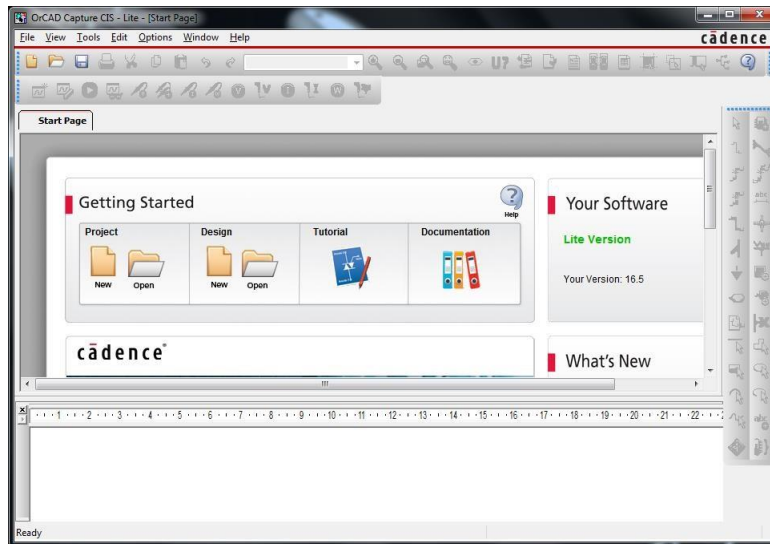
2 Creating a new project

1. The following directory should have appeared in your Start menu:



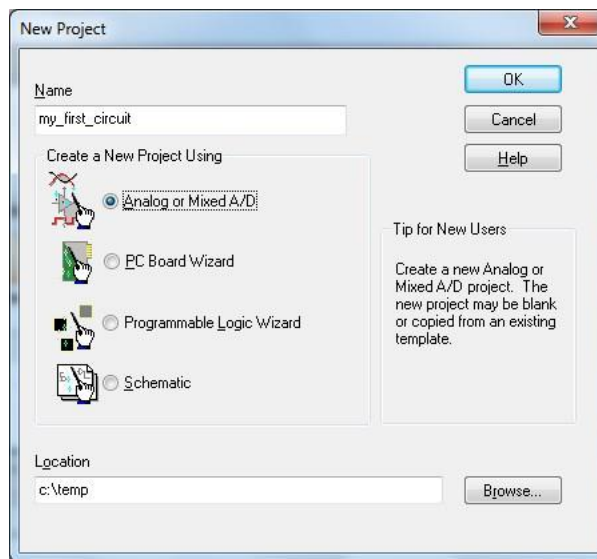
Launch the software by clicking on OrCAD Capture CIS Lite as highlighted above.

2. The following window will be displayed:

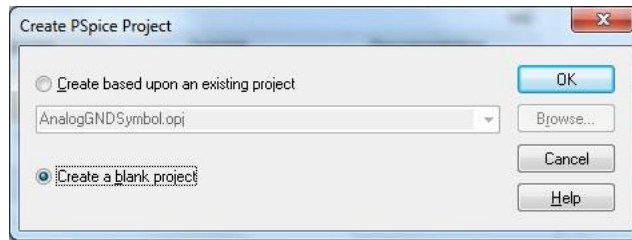


Create a new document by clicking on *File* → *New* → *Project* or on the icon in the top-left of the screen.

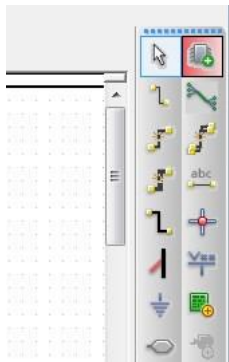
3. In the subsequent window, give your circuit a name, assign a location for its files to be saved, and select Analog or Mixed A/D as its type. Then click *OK*.



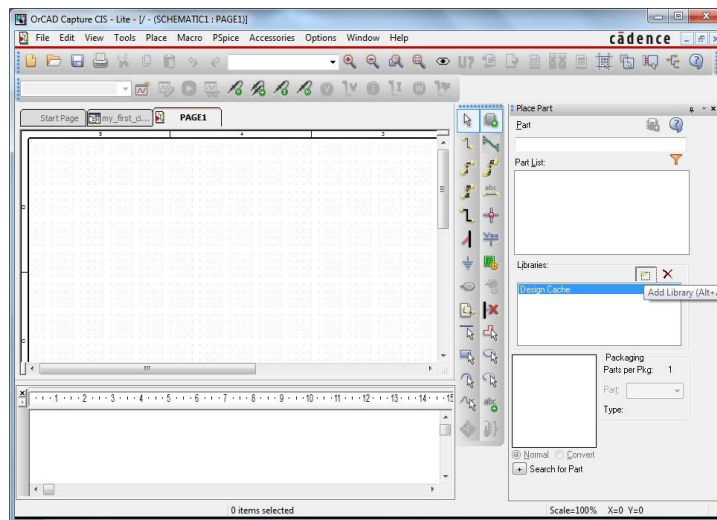
In the window which follows this, select Create a blank project and click *OK*.



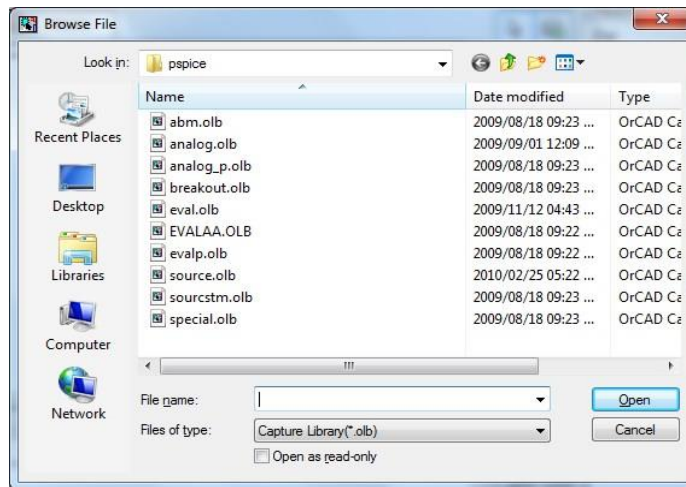
4. You will then be presented with a blank schematic. Before you can place components on your schematic, OrCAD needs libraries from which to take these components. Click on the Place part icon on the top right of the screen.



A window will appear on the right hand side of the screen. In it, click on Add library which can be found towards the middle of the right-hand side.

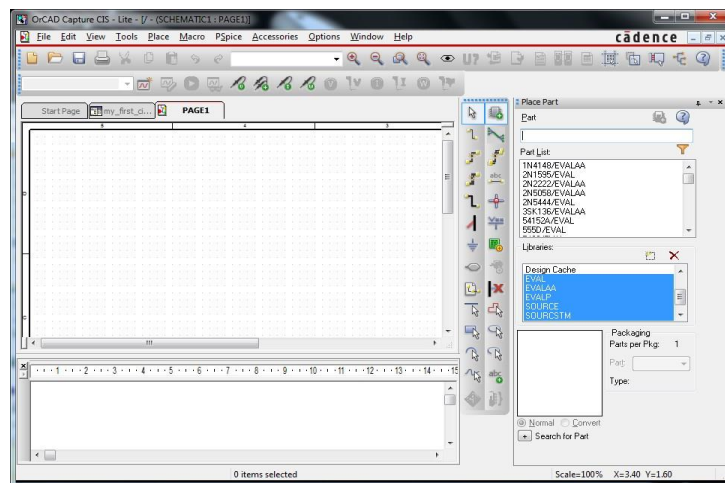


5. A window should open, showing you the contents of a directory named pspice. In the event that this does not happen, please navigate to C:\OrCAD\OrCAD .16.5 Lite\tools\capture\library\pspice or similar location in your install directory, so that you see the following library files:



Select all the files, then click Open.

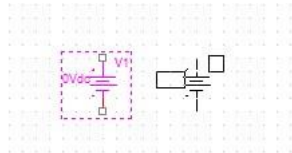
6. The boxes on the right-hand side of your screen will now be populated with names of circuit components:



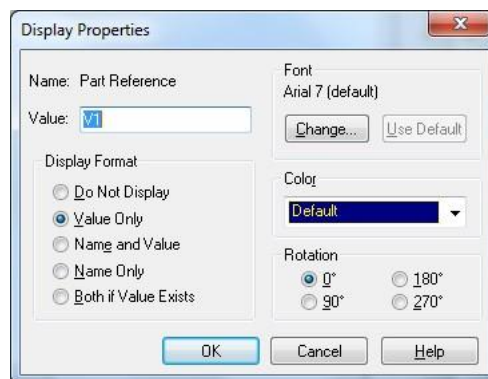
You are now ready to begin building a circuit schematic.

3 Creating a simple circuit

1. We will construct and simulate a simple circuit similar to that required in the pre-prac assignment. In the box labelled Part type vdc in order to place a DC voltage source. An item labelled VDC/SOURCE will be highlighted. Double-click on it and place the DC voltage source somewhere on your schematic.

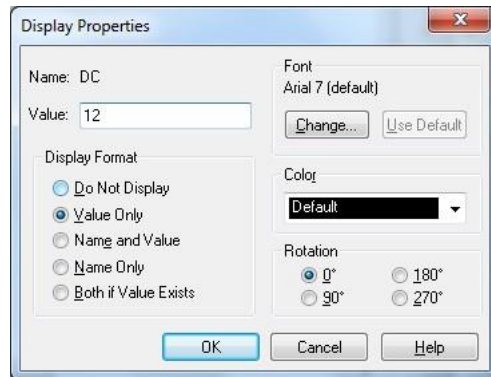


2. Double-click on the label V1



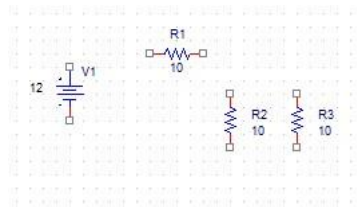
Should you desire (and this is often an advantage in more complicated circuits) you may rename your voltage source. If you rename it, remember its name! It will be used later.

3. Double-click the label 0Vdc. A similar window will appear. Here you assign a DC voltage to the source. Assign 12 V by typing 12 into the box as shown:



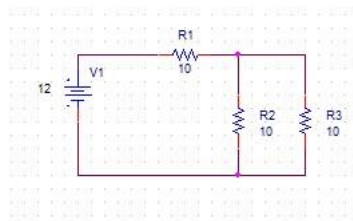
Note: Do not type 12V or any other combination of letters, simply the number 12 will suffice. Once you are done, click OK.

4. Place resistors by typing r into the Part box. The item R/ANALOG will be highlighted, double-click it to place resistors. Place three resistors and reassign each of them to have a resistance of 10 (in exactly the same way we assigned a voltage value to the voltage source in the previous step) as shown:



Note: You can rotate a resistor by clicking on it to select it, then pressing R on your keyboard while it is selected. Note: The default value of 1k with which the resistors were created means 1 kilo ohm, or 1000 . OrCAD will accept any SI postfix to modify numerical values in this manner.

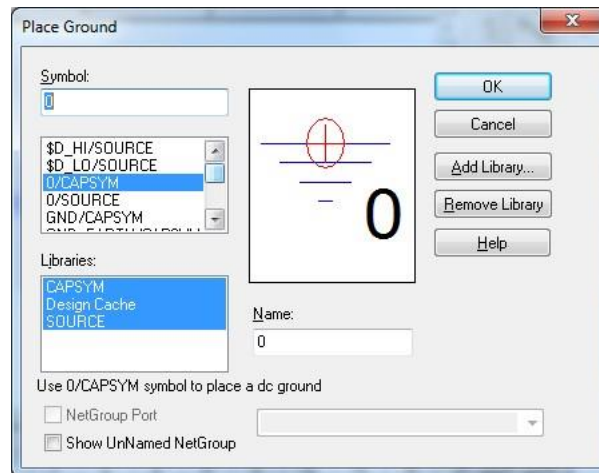
5. Once you have placed the resistors, press Esc. on your keyboard to make sure that your cursor is clear, then press W to begin placing wires. Place wires by clicking once where the wire should start, then once where it should end. Connect the components you have placed in the following manner:



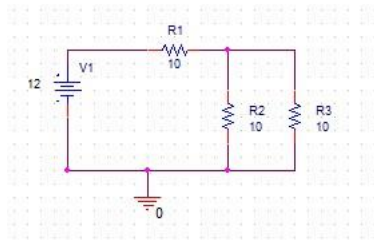
6. Before the circuit can be simulated, a ground reference is required. You can add a ground by clicking on the Place ground button to the right of your screen:



The following window will appear:

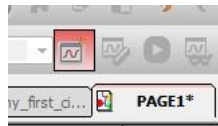


Double-click in 0/CAPSYM and place the ground somewhere appropriate on your schematic. Connect it to the rest of the circuit with a wire as in the previous steps.

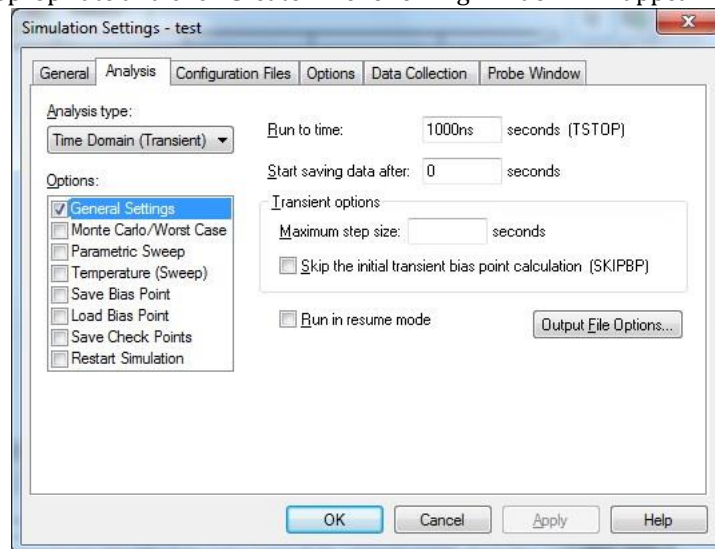


4 Simulating a Circuit (DC Bias Point analysis)

1. Your circuit is now ready to simulate! The first step is to create a PSpicesimulation profile. The button is located near the top of the window:



In the window that follows, give the simulation profile any name you feel is appropriate and click Create. The following window will appear:



Note: In some instances, the window might appear minimised. You can bring it to the front using the taskbar.

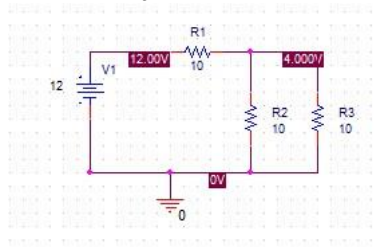
2. Under Analysis type: select Bias Point. Then click OK.

3. Start the simulation by clicking the run button:

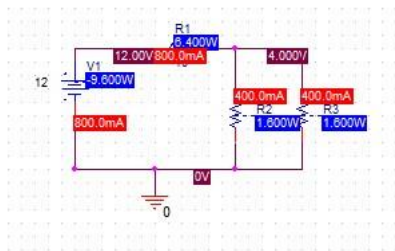


A warning will pop up, informing you that changes will be saved and the undo/redo functionality will not be available after this point. This is normal. Click Yes to proceed with the simulation.

4. A PSpice window will appear, but we have not asked PSpice to graph anything, so there will be no graph in it. Rather the DC bias point of your circuit will be shown on your schematic:



By default, only the node voltages (with reference to ground) are shown at each node. You can also show currents in each branch and power dissipated or provided by each circuit element by clicking on the appropriate buttons.



5. The buttons just to the left of these are for placing probes.

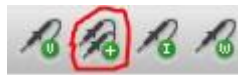


These probes measure node voltage, voltage with a different reference, current and power respectively and are not particularly useful with a bias-point simulation. They are however useful with the other kinds of simulations.

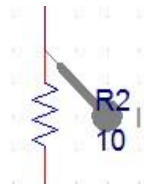
5 Simulating a Circuit (DC Sweep)

In your practical, you will be required to use the DC Sweep simulation. This type of simulation 'sweeps' one value, such as the input voltage, through a range of specified values and measures the change in another set of values. Let say we want to measure the current through R_2 and the voltage across R_1 (from the source to ground) as V_1 is changed from 0V to 12V in steps of 0.5V.

1. First we start by placing the measurement components. Start by clicking on the current probe button.



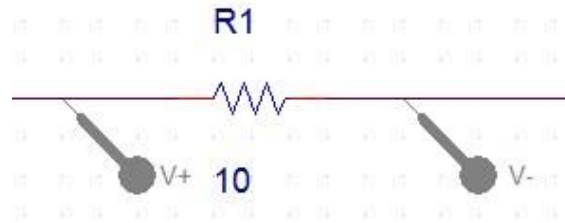
2. Place it between the top of R_2 and the wire (between the brown/maroon parts and the red part of the resistor)



3. Now select the voltage probes to measure across a component.



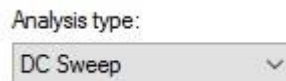
4. Place the positive terminal to the left of R_1 and the negative terminal to the right. Note that the first click places the positive terminal and the second click the negative.



5. With the measurement points selected we can start setting up the simulation. Click on the edit simulation button. A new window should open but it might be hidden in your taskbar.



6. Under *Analysis Type* select DC Sweep.



7. Fill in the highlighted fields as shown in the screenshot below. As requested by the question, V_1 is swept from 0 to 12 in increments of 0.5. Click the OK button.

Sweep variable

☒ Voltage source
☐ Current source
☐ Global parameter
☐ Model parameter
☐ Temperature

Name:
Model type:
Model name:
Parameter name:

Sweep type

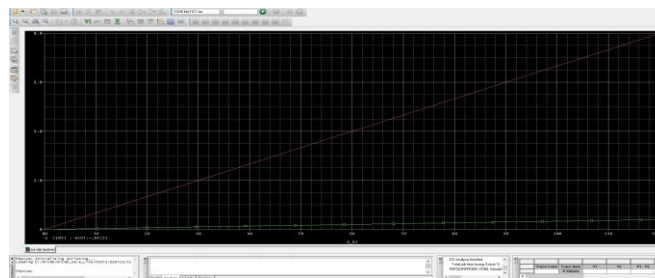
☒ Linear
☐ Logarithmic
☐ Value list

Decade
Start value:
End value:
Increment:

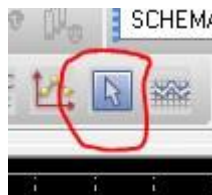
8. Now click on the Run PSpice button, shown below



9. A new window should be created (again, it may be hidden in the taskbar). It should look similar to the one below. If you look to the bottom left of the graph you will be a legend indicating what each line (called traces) measures.



10. Now click on the cursor, shown below. Two lines (x and y) should appear on your graph). You can drag it on the graph or click on a point to measure a specific value. On the bottom right there is a table which you can use with the cursor to read specific values.



11. Finally note that there is a bug in OrCAD 16.6 where you cannot read the last interval of your simulation using the cursor. In the case of this example, you cannot drag the cursor beyond 11.5 on the x-axis. If you need a value beyond this point, say a reading at the maximum of the graph, you can still read it using the buttons shown below (the fifth button instantly takes you to the maximum).

